

# BIOL-1 Practice Test 05

## Comprehensive Final Review (Modules 01–16)

College of the Redwoods — BIOL-1: General Biology (Pelican Bay, Fall 2026)

**Instructions:** This practice test samples **big ideas** from every module. Work through Parts A–C. For Part C, answer in complete sentences.

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### Part A: Multiple Choice (75 questions)

*Choose the best answer for each question.*

#### Module 01: Study of Life

1. Which property is shared by every living organism in this course's working definition of life?
  - A) They must move from place to place
  - B) They all use photosynthesis
  - C) They are composed of one or more cells
  - D) They are all multicellular
  
2. In the scientific method, a testable, proposed explanation that follows from observations is best called a:
  - A) Theory
  - B) Hypothesis
  - C) Law
  - D) Conclusion
  
3. In a controlled experiment, the factor the researcher deliberately changes is usually the:
  - A) Dependent variable
  - B) Confounding variable

C) Independent variable

D) Control group only

4. The maintenance of a stable internal environment (for example, temperature or blood sugar) despite external change is called:

A) Differentiation

B) Homeostasis

C) Metabolism

D) Evolution

5. The simplest level in the hierarchy “cell → tissue → organ → organ system → organism” is the:

A) Cell

B) Organ

C) Tissue

D) Population

## Module 02: Basic Chemistry

6. The **atomic number** of an element equals the number of:

A) Neutrons only

B) Electrons only in ions

C) Protons in the nucleus

D) Isotopes

7. Atoms of the same element that differ in neutron number are called:

A) Ions

B) Molecules

C) Isotopes

D) Polymers

8. A bond formed mainly by **transfer** of electrons from one atom to another, producing attraction between opposite charges, is:

- A) Covalent
- B) Hydrogen
- C) Van der Waals
- D) Ionic

9. Water is polar mainly because:

- A) It has hydrogen atoms only
- B) It cannot hydrogen-bond
- C) Shared electrons are pulled unevenly, creating partial charges
- D) It always has pH 7

10. A solution with a **lower** pH than 7 is:

- A) Basic
- B) Acidic
- C) Neutral
- D) A solution with pH above 7

### **Module 03: Organic Molecules**

11. The element that forms the backbone of biological macromolecules is:

- A) Carbon
- B) Sodium
- C) Nitrogen
- D) Iron

12. Amino acids link by **dehydration synthesis** to form proteins, releasing:

- A) CO<sub>2</sub>
- B) O<sub>2</sub>
- C) ATP
- D) H<sub>2</sub>O

13. The monomer (building block) of proteins is the:

- A) Monosaccharide
- B) Nucleotide
- C) Fatty acid
- D) Amino acid

**14.** Starch and glycogen are storage forms of which class of macromolecule?

- A) Lipid
- B) Carbohydrate
- C) Protein
- D) Nucleic acid

**15.** RNA differs from DNA in several ways; one common contrast is that RNA uses the base **uracil** instead of:

- A) Thymine
- B) Adenine
- C) Cytosine
- D) Guanine

#### **Module 04: Cells**

**16.** A central idea of cell theory is that **new cells** arise from:

- A) Pre-existing cells
- B) Spontaneous generation from broth
- C) Only the ER
- D) Viruses

**17.** Prokaryotic cells lack which structure that eukaryotic cells usually have?

- A) Ribosomes
- B) DNA
- C) A membrane-bound nucleus
- D) A plasma membrane

**18.** Most ATP for a typical animal cell is generated in the:

- A) Nucleus
- B) Mitochondria
- C) Lysosome
- D) Centriole

**19.** In plant cells but not typical animal cells, you expect to find:

- A) A cell wall outside the membrane
- B) Mitochondria
- C) Ribosomes
- D) A nucleus

**20.** Proteins are assembled on:

- A) Ribosomes
- B) The Golgi apparatus
- C) The smooth ER only
- D) Peroxisomes only

### **Module 05: Membranes**

**21.** The basic structure of the plasma membrane is described as a:

- A) Single layer of proteins
- B) Fluid mosaic of lipids and proteins
- C) Static rigid wall
- D) DNA bilayer

**22.** Simple diffusion moves molecules:

- A) Only against a gradient with ATP
- B) Only in plants
- C) Only as ions
- D) With their concentration gradient, no ATP required for the diffusion step

**23.** In osmosis, **water** moves:

- A) Against gradients always
- B) Toward the area with lower water (higher solute) if the membrane allows water through
- C) Only into dead cells
- D) Only when sodium ions move first

**24.** Active transport differs from simple diffusion because it:

- A) Never uses proteins
- B) Can move substances against their concentration gradient using cellular energy
- C) Only moves water
- D) Does not depend on membrane proteins

**25.** Which molecules typically cross a pure lipid bilayer **least** easily without help?

- A) Small nonpolar gases
- B) Water (tiny, though polar)
- C) Large, charged ions without channels/carriers
- D) Small nonpolar molecules

### **Module 06: Metabolism**

**26.** The molecule often called the cell's immediate **energy currency** is:

- A) DNA
- B) ATP
- C) Glucose only
- D) mRNA

**27.** Enzymes speed reactions mainly by:

- A) Lowering activation energy
- B) Raising activation energy
- C) Destroying products
- D) Removing all substrates

**28.** The summary outcome of **aerobic cellular respiration** (big picture) is that glucose is broken down and much energy is captured as:

- A) Only heat with no ATP
- B) ATP (and heat), with CO<sub>2</sub> and water among outputs
- C) Light
- D) Only lipids

**29.** The light-capturing reactions of photosynthesis happen in chloroplast structures associated with:

- A) Mitochondrial cristae
- B) Lysosomes
- C) Thylakoids / grana (chloroplast membranes)
- D) The nucleolus

**30.** Fermentation is a way some cells continue **glycolysis** when:

- A) Oxygen is abundant
- B) Photosynthesis is maximal
- C) Oxygen is limited, regenerating NAD<sup>+</sup> so glycolysis can keep running
- D) DNA replicates

### **Module 07: Molecular Genetics**

**31.** The phrase **central dogma** most often summarizes:

- A) DNA → RNA → protein in typical gene expression
- B) Protein → RNA → DNA
- C) Only replication
- D) Mendel's laws only

**32.** In DNA, adenine pairs (base-pairs) with:

- A) Guanine
- B) Cytosine
- C) Uracil
- D) Thymine

**33. Transcription** produces:

- A) An RNA copy of a gene, using DNA as a template
- B) A new DNA double helix only
- C) A finished protein directly
- D) Only tRNA with no mRNA

**34. Translation** reads which molecule to build a protein?

- A) DNA directly in the nucleus only
- B) Only rRNA with no codons
- C) mRNA at the ribosome
- D) Lipid bilayers

**35.** A change in the **DNA sequence** that may alter an encoded protein is a:

- A) Epigenetic mark only (same textbook definition)
- B) Mutation
- C) Gene flow event
- D) Bottleneck only

#### **Module 08: Cellular Genetics**

**36.** Somatic cells reproduce for growth and repair mainly by:

- A) Meiosis
- B) Fermentation
- C) Mitosis (exact copy distribution to two daughter cells)
- D) Translation

**37.** Meiosis produces gametes with:

- A) The same chromosome number as the parent body cell
- B) No chromosomes
- C) Only mitochondrial DNA
- D) Half the chromosome number (haploid from diploid), with genetic shuffling

**38.** Crossing over (exchange between homologous chromosomes) increases variation mainly during:



- A) Mitosis only
- B) Translation
- C) DNA replication only in skin
- D) Prophase I of meiosis

**39.** A diploid human body cell has **46** chromosomes; a normal human sperm has about:

- A) 46
- B) 23
- C) 92
- D) 0

**40.** The term **somatic** cell refers to:

- A) Body cells that are not gametes
- B) Only gametes
- C) Bacterial cells
- D) Dead cells only

### **Module 09: Inheritance Genetics**

**41.** An organism with two identical alleles for a gene is:

- A) Heterozygous
- B) Homozygous
- C) Polyploid only
- D) Always dominant

**42.** In simple dominance, if **T** is tall and **t** is short, a **Tt** plant's phenotype is usually:

- A) Short
- B) Medium always
- C) Lethal
- D) Tall (dominant trait expressed)

**43.** A **Punnett square** is used to predict:

- A) Offspring genotypes / phenotypes from parental gametes
- B) Mitotic spindle angles only
- C) Only mutation rates
- D) Ecological carrying capacity

**44.** A pedigree chart tracks:

- A) Only fossils
- B) Traits through family generations
- C) Only food webs
- D) Only soil nutrient cycles

**45.** Mendel's idea that "heritable factors" (alleles) **segregate** into gametes underpins:

- A) Only natural selection
- B) The law of segregation (intro level)
- C) Only photosynthesis
- D) Binomial nomenclature only

### **Module 10: Epigenetics**

**46. Epigenetics** in this course most closely means:

- A) Changing the DNA letter sequence permanently by CRISPR only
- B) Random gamete fusion
- C) Fermentation pathways
- D) Regulation of gene activity without changing the DNA sequence

**47. DNA methylation** that silences a gene region is a textbook example of:

- A) A missense mutation
- B) Mitosis
- C) An epigenetic mark (often "off")
- D) Independent assortment

**48.** X-inactivation in female mammals (one X compacted in each cell) helps explain, for example:

- A) Why men have Y in every liver cell only
- B) Why plants lack DNA
- C) Bacterial transformation only
- D) Calico cat patch color patterns (classic intro story)

**49.** Compared with a **mutation**, a typical epigenetic change to gene activity:

- A) Always changes the codon sequence
- B) Cannot respond to environment
- C) Is often reversible at the “volume knob” level of expression
- D) Only happens in viruses

**50. Gene expression** (big picture) refers to:

- A) Only DNA replication
- B) A gene being used to produce RNA and often protein
- C) Only photosynthesis
- D) Only drift

## **Module 11: Genomics & Biotechnology**

**51. PCR** is used in teaching examples to:

- A) Amplify a targeted DNA segment in a tube
- B) Photograph mice only
- C) Replace meiosis
- D) Create ATP from sunlight

**52.** Gel electrophoresis separates DNA fragments mainly because:

- A) The phosphate backbone makes DNA negatively charged; fragments move through a gel toward positive
- B) Heat destroys all DNA the same way
- C) Lipids are charged like DNA
- D) Only RNA enters the gel

**53. A transgenic** organism carries:

- A) No DNA
- B) Only mitochondrial DNA from mother
- C) Foreign DNA introduced into its genome (GMO framing)
- D) Only fossil genes

**54. CRISPR** is often introduced as a tool to:

- A) Edit DNA at targeted sites (with big ethics questions)
- B) Ignore genes
- C) Make fossils
- D) Measure pH only

**55. A restriction enzyme** cuts DNA at:

- A) Random weather patterns
- B) Only RNA
- C) Only proteins
- D) Specific recognition sequences

## **Module 12: Darwin & Evolution**

**56. Natural selection** requires heritable variation, differential reproduction, and:

- A) Goal-directed “wants” of species
- B) Lamarckian inheritance of acquired traits as the main mechanism
- C) Perfect environments only
- D) Overproduction of offspring relative to resources (broad textbook framing)

**57. Descent with modification** means:

- A) Each species is fixed forever
- B) Only bacteria evolve
- C) Lineages change over time; related species share ancestors with later differences
- D) Evolution cannot be tested

**58. Homology** (for example similar limb bone layout, different functions) supports:

- A) Identical environments only
- B) No shared history
- C) Common ancestry with adaptive change
- D) Only convergent evolution in every case

**59. Fitness** in evolution means roughly:

- A) Reproductive success / allele contribution to the next generation
- B) Bench press strength only
- C) Survival with zero offspring forever
- D) Size only

**60.** Darwin emphasized that **populations** evolve rather than individuals mainly because:

- A) Individuals rewrite DNA by thinking
- B) Fossils are not real
- C) Allele frequencies change across generations in a population
- D) Only one individual matters per species

### **Module 13: How Populations Evolve**

**61. Microevolution** is defined in this course as:

- A) An individual growing taller in one year
- B) Instant new phylum in one birth
- C) Only mutation in skin cells with no offspring
- D) Change in allele frequencies in a population over time

**62.** The **Hardy–Weinberg** model is mainly useful as:

- A) A null / baseline to compare real allele-frequency change against
- B) Proof that evolution always speeds up
- C) A ban on mutation
- D) A species concept

**63.** The **founder effect** is a type of:

- A) Directional selection only
- B) Gene flow from infinite migrants always
- C) Genetic drift involving a small founding group
- D) Photosynthesis error

**64. A bottleneck** reduces allele diversity mainly because:

- A) Selection cannot happen
- B) Mutation stops
- C) K always equals zero
- D) A sudden severe population crash makes survivors' alleles unrepresentative by chance

**65. Stabilizing selection** on a trait tends to:

- A) Favor both extremes
- B) Always eliminate all variation instantly
- C) Require polyploidy
- D) Favor the middle; select against extremes

#### **Module 14: Macroevolution**

**66. The biological species concept** emphasizes:

- A) Reproductive isolation between groups in nature (big-picture framing)
- B) Color only
- C) Cell size only
- D) Fossil size only

**67. Different firefly flash patterns that reduce mating attempts are best described as:**

- A) A postzygotic barrier (hybrid infertility)
- B) Geographic separation with no mating cues
- C) Sympatric polyploidy in one generation
- D) A prezygotic behavioral barrier to reproduction

**68. Mule sterility (horse × donkey hybrid) is a classic example of:**

- A) Prezygotic temporal isolation between species
- B) A postzygotic barrier (reduced hybrid fertility)
- C) Convergent evolution in ungulates
- D) A prezygotic gametic barrier only

**69. Allopatric** speciation often begins with:

- A) Geographic separation of formerly connected populations
- B) Same habitat, no separation
- C) Only sympatry by definition
- D) Only hybrid swarms always

**70. Sympatric** speciation examples in intro courses often highlight:

- A) Only continental drift with zero genetics
- B) Plant polyploidy in the same region as a parent type
- C) Only fossils
- D) Only density independence

### **Module 15: Population & Systems Ecology**

**71.** Early **exponential** population growth (unlimited-resource phase) is often **J**-shaped because:

- A) Growth rate is near zero always
- B) Carrying capacity is always zero
- C) Per-capita growth can stay high while numbers are still far below limits
- D) Only density-dependent rules apply on day one

**72. Carrying capacity (K)** in a logistic model is:

- A) The mutation rate
- B) Always infinite
- C) Gene flow only
- D) A long-run ceiling set by resources and other limits

**73.** A disease spreading **more easily in crowded** herds is often modeled as:

- A) Density-dependent regulation
- B) Density-independent
- C) Only abiotic freeze
- D) Carrying capacity with no biology

74. The rough ~10% **rule** between trophic levels implies:

- A) 100% of energy becomes flesh always
- B) Only a small fraction of stored energy transfers to the next consumer level (lots lost as heat/work)
- C) Decomposers violate thermodynamics
- D) Producers get energy from consumers' heat cycling perfectly

75. In ecosystems, **matter** (nutrient elements) cycles while energy from sunlight typically:

- A) Is recycled exactly like nitrogen atoms with no loss
- B) Stays only in rocks forever with zero biology
- C) Doubles at every trophic level
- D) Flows through and ends largely as heat (not recycled like chemical elements in the same way energy is framed)

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## Part B: Fill in the Blank (30 questions)

*Write a short term or phrase in each blank (usually one or two words).*

### Module 01

76. In a controlled experiment, a group that does **not** receive the experimental treatment (used for comparison) is often called the

\_\_\_\_\_ group.

77. All living things must maintain organization, reproduce, respond to stimuli, undergo metabolism, and evolve as populations; many also grow and develop while maintaining

\_\_\_\_\_.



## Module 02

78. A covalent bond involves **sharing** pairs of \_\_\_\_\_ between atoms.

79. A neutral atom has equal numbers of protons and \_\_\_\_\_ (in the standard model for this course).

## Module 03

80. The chemical bond that links amino acids in a chain is a \_\_\_\_\_ bond.

81. Fats and oils belong to the macromolecule class called \_\_\_\_\_.

## Module 04

82. Membrane-bound sacs of digestive enzymes that break down macromolecules in many eukaryotic cells are \_\_\_\_\_

83. The control center that holds most DNA in a eukaryotic cell is the \_\_\_\_\_.

## Module 05

84. Transport proteins that open the door for molecules **without** using ATP for that step (along a gradient) perform \_\_\_\_\_ transport.

85. The phospholipid tails of the bilayer are \_\_\_\_\_  
(hydrophilic or hydrophobic)—choose the correct term.

## Module 06

86. The anaerobic portion of glucose breakdown that happens in the cytoplasm (and yields a little ATP and pyruvate) is

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87. The pigment that captures light energy in plants and many photosynthetic organisms is

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## Module 07

88. During DNA replication, the enzyme class that **matches** incoming nucleotides to the template strand (5'→3' synthesis on a daughter strand) is

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polymerase (family name only is fine).

89. A **three**-nucleotide mRNA code that specifies an amino acid (or stop) is a

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## Module 08

90. The stage of mitosis where **sister chromatids** are pulled apart toward opposite poles is

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91. Homologous chromosomes pair up during

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I of meiosis.

## Module 09

92. An observable trait (tall, short, etc.) is called the

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93. When both alleles are visibly expressed in the heterozygote (classic AB blood-type framing), inheritance is often called

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## Module 10

94. Methyl groups added to DNA can often **silence** nearby genes as an

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mark (adjective form used in class).

95. Proteins that help **turn transcription on or off** near promoters are often called transcription

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## Module 11

96. A **plasmid** is a small, circular **extra** piece of DNA often used in bacterial \_\_\_\_\_ work (one-word lab context is fine).

97. To cut and paste a gene, labs often use **restriction** enzymes and

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enzyme to seal DNA back together.

## Module 12

98. Heritable differences among individuals in a population are called

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99. When unrelated lineages evolve **similar** traits because of similar selection pressures (not shared recent ancestry), it is

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evolution.

## Module 13

100. When chance events change allele frequencies (especially in small populations), the umbrella term is genetic

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**101.** If **two** alleles A and a have frequencies **p** and **q** in a diploid population at Hardy–Weinberg equilibrium, the expected heterozygote frequency is \_\_\_\_\_ (symbolic answer).

## Module 14

**102.** Barriers that block mating or fertilization **before** a zygote forms are \_\_\_\_\_ barriers.

**103.** A **reduced hybrid** survival or fertility after a zygote forms is a \_\_\_\_\_ zygotic barrier category.

## Module 15

**104.** All the organisms of one species living in an area at one time form a(n) \_\_\_\_\_

**105.** Organisms that make organic matter from inorganic carbon (often via photosynthesis) are **primary** \_\_\_\_\_ (trophic role).

## Part C: Free Response (30 questions)

*Answer in complete sentences. Each item is tagged with its home module.*

**106. (Module 01)** State two characteristics of life and **briefly** explain each in your own words.

**107. (Module 01)** What is the difference between a **scientific theory** and a **hypothesis** in everyday lab language?

**108. (Module 02)** Explain how water's polarity helps it dissolve **ionic** compounds such as table salt at the molecular level (big picture, no deep math).

**109. (Module 02)** Why does a **buffer** matter in blood and cellular chemistry (one or two sentences of “what it does”)?

**110. (Module 03)** Name the **four** major classes of biological macromolecules and, for **one** class you choose, name its monomer type.

**111. (Module 03)** What happens to a protein's function when it is **denatured**, and what is one common cause of denaturation?

**112. (Module 04)** Compare **prokaryotic** and **eukaryotic** cells in **two** sentences (nucleus + relative complexity).

**113. (Module 04)** What is the main job of the **mitochondria**, and where in the cell does it do that work?

**114. (Module 05)** Contrast **passive** and **active** transport across a membrane with **one example** of each.

**115. (Module 05)** In plain language, what happens to a cell placed in a **very salty** external solution if the membrane is permeable to water only (animal cell framing)?

**116. (Module 06)** Write one sentence linking **photosynthesis** and **cellular respiration** as complementary carbon/energy themes on Earth.

**117. (Module 06)** Why do enzymes stop speeding a reaction if the temperature becomes **too high**?

**118. (Module 07)** Describe **transcription** and **translation** in **two** short sentences, using the words **DNA**, **RNA**, **ribosome**, and **protein** sensibly.

**119. (Module 07)** Why is **DNA replication** described as **semiconservative** at the big-picture level?

**120. (Module 08)** Contrast the **goal** of mitosis versus meiosis in a multicellular animal (what each type produces).

**121. (Module 08)** Name **two** ways meiosis increases **genetic variation** besides random fertilization.



**122. (Module 09)** In a monohybrid cross  $Aa \times Aa$  with complete dominance, what are the expected **phenotype** ratios in offspring?

**123. (Module 09)** Define **heterozygous** and give a **letter example** (any gene).

**124. (Module 10)** In one short paragraph, explain how **epigenetics** differs from a **mutation**.

**125. (Module 10)** Give one **realistic** example of how environment **without** changing DNA letters might still influence **trait expression** at the gene-regulation level.

**126. (Module 11)** In two sentences, what is a **GMO** in food/ag contexts, and why do people disagree ethically?

**127. (Module 11)** What does **gel electrophoresis** show a biologist about DNA samples (smaller vs larger fragments)?

**128. (Module 12)** Explain **natural selection** in four short sentences aimed at a friend who has never taken biology.

**129. (Module 12)** Name **two** independent lines of **evidence** that life shares common ancestry and **briefly** say what each shows.

**130. (Module 13)** Define **microevolution** and give **one** example of a mechanism that can change allele frequencies **other** than natural selection.

**131. (Module 13)** Contrast **directional**, **stabilizing**, and **disruptive** selection in one sentence **each**.

**132. (Module 14)** Contrast **allopatric** and **sympatric** speciation with **one sentence each**.

**133. (Module 14)** Give **one** example of a **prezygotic** barrier and **one** of a **postzygotic** barrier.

**134. (Module 15)** Contrast **exponential** vs **logistic** population growth; mention **carrying capacity** in your answer.

**135. (Module 15)** Explain the **10% energy rule** and **why** long food chains support fewer top predators than short chains from the same plant base.

**136. (Module 16)** In one paragraph, connect **DNA or genes**, **natural selection**, and **ecosystem limits** in a single biological system.

**137. (Module 16)** Explain why a strong biology answer often needs to name the **scale** of the evidence: molecule, cell, organism, population, community, or ecosystem.

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*End of Practice Test 05*